

Sankey Plot Activity Submission

Experiment: Career Pathway Analysis using Sankey Plot (R - networkD3)

R Code

```
packages <- c("networkD3", "htmlwidgets")

for (pkg in packages) {
  if (!require(pkg, character.only = TRUE)) {
    install.packages(pkg, repos = "https://cloud.r-project.org")
    library(pkg, character.only = TRUE)
  }
}

nodes <- data.frame(
  name = c(
    "Computer Science",
    "Information Technology",
    "Electronics",
    "Mechanical",
    "Civil",
    "AI & ML",
    "Data Science",
    "Cyber Security",
    "Embedded Systems",
    "Manufacturing",
    "Construction",
    "Software Engineer",
    "Data Analyst",
```

```
      "Research",  
      "Higher Studies"  
    ),  
    stringsAsFactors = FALSE  
  )
```

```
links <- data.frame(  
  source = c(  
    0,0,0,  
    1,1,  
    2,  
    3,  
    4,  
    5,5,  
    6,6,6,  
    7,7,  
    8,8,  
    9,  
    10,10  
  ),  
  target = c(  
    5,6,7,  
    6,7,  
    8,  
    9,  
    10,  
    11,13,  
    11,12,14,  
    11,13,  
  )  
)
```

```

      13,14,
      11,
      11,14
    ),
    value = c(
      80,70,50,
      60,50,
      90,
      100,
      80,
      50,30,
      40,60,30,
      35,25,
      40,50,
      70,
      50,30
    )
  )
)

p <- sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 14,
  nodeWidth = 35
)

```

```
saveWidget(p, "sankey_plot.html", selfcontained = TRUE)
```

```
cat("Sankey plot saved as sankey_plot.html\n")
```

```
browseURL("sankey_plot.html")
```

Question 3 - Interpretation

- Specialization receiving the highest number of students: Data Science (130 students)
- Career outcome with the largest overall inflow: Software Engineer (245 students)
- Department contributing the most students to Software Engineer positions: Computer Science

Question 4

A Sankey diagram is more appropriate because it shows the flow of students across multiple stages (Departments → Specializations → Career Outcomes). The width of each link represents the number of students, making major pathways easy to identify. Bar charts and pie charts cannot effectively show these flows.

Question 5

1. Use different colors for each department.
2. Add interactive tooltips displaying the exact number of students for each flow.

Output

